

Building an Interactive Electrostatics Simulator: An AI-Assisted Recreation of Charges, Fields, and Parallel Plate Capacitors

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Abstract

In this project, we developed an interactive simulation to demonstrate the Physics II concepts of electric fields and electric potential. The simulation allows users to place and move positive and negative charges, observe electric field lines and field vectors, measure electric field strength and electric potential, trace equipotential lines, and explore a parallel plate capacitor. Throughout the project, we evaluated the strengths and limitations of the simulation and discussed possible improvements, such as adding guided tutorials, more examples, and additional features. Overall, our simulation provides a hands-on way to better understand electrostatic concepts that are often difficult to visualize using equations alone.

Introduction

For our project, we chose the Physics II concept of electric fields and electric potential. These concepts explain how charged objects interact and how they create forces on other charged particles. To better understand these ideas, we used the Charges & Fields simulation, which allows us to place positive and negative charges, move them around, and observe how the electric field lines and electric potential change. The simulation also includes a parallel plate capacitor mode that demonstrates how a nearly uniform electric field forms between two oppositely charged plates.

We found the simulation to be effective because it helped us visualize concepts that can be difficult to understand from equations alone. Being able to move the charges and instantly see how the electric field and potential changed made it easier to connect the theory from class with what was happening in the simulation. We think the simulation could be improved by adding more explanations and guided activities that describe why the electric field changes and challenge students to predict the results before making changes. Overall, this simulation provided a hands-on way to strengthen our understanding of electric fields and electric potential.

Methods

For this project, we used the web interface of Claude (Anthropic), specifically the Claude Sonnet 5 model, on July 20, 2026, and no other AI tools were utilized. The simulation was built iteratively over six prompts. We started with an initial request to build a simulation modeled on PhET's "Charges and Fields," confirmed no changes were needed, and asked where to find the file. Next, we requested the addition of a parallel plate capacitor mode and asked for the entire project to be re-exported as a standalone HTML file for portal compatibility. Finally, we requested field lines, a live field and potential sensor, and a test charge showing force and acceleration as a vector, with each step building on the previous one rather than starting over. I verified the simulation by interacting with it directly—dragging charges and confirming that the field vectors, field lines, and equipotential contours behaved correctly, such as field lines running from positive to negative charges and a roughly uniform field between the capacitor plates. While we did not verify every line of the underlying algorithms, like the contour-tracing math, we ensured the on-screen physics behavior was accurate. A major challenge was that the first version was a React component that wouldn't run standalone, requiring a rewrite into a single, self-contained HTML file. Additionally, equipotential lines needed a contour-tracing method rather than a simple formula once multiple charges and plates were involved, and performance tuning was necessary to maintain smoothness while dragging charges. Ultimately, while Claude quickly handled the physics-simulation boilerplate—such as vector math, canvas rendering, drag interactions, and contour tracing—human judgment was still essential to select appropriate features, fix format mismatches with the assignment requirements, and confirm the simulation accurately reflected real electrostatics rather than just looking plausible.

Results

Our simulation was designed to help users explore the concepts of electric fields and electric potential through an interactive interface. The simulation includes two main modes: Point Charges and Parallel Plate Capacitor. In the Point Charges mode, users can add positive and negative charges, move them around the screen, and immediately observe how the electric field lines and field vectors change. A movable sensor displays the electric potential (V), electric field magnitude ($|E|$), and the x- and y-components of the electric field at any location. Users can also trace equipotential lines and place a test charge to visualize the direction and relative strength of the electric force.

The Parallel Plate Capacitor mode demonstrates how a nearly uniform electric field forms between two oppositely charged plates. Users can adjust the distance between the plates, change the charge magnitude, and even reverse the polarity to observe how these changes affect the electric field. Throughout the simulation, the graphics update in real time, allowing users to directly connect the Physics II concepts with their visual representations. Figures 1 and 2 show the Point Charges and Parallel Plate Capacitor modes used in our project. Overall, the simulation provides an interactive way to visualize electrostatic principles that are often difficult to understand through equations alone.

Discussion

One of the biggest strengths of our simulation is how interactive it is. Instead of only looking at diagrams or solving equations, we were able to move charges around and instantly see how the electric field lines, field vectors, and electric potential changed. The sensor and test charge also made it easier to understand what was happening because they displayed real-time values for the electric field and potential at different locations. This helped us connect the concepts we learned in Physics II with what we were seeing in the simulation.

Our simulation also has some limitations. It only models ideal situations, so it does not include real-world effects like dielectric materials, three-dimensional electric fields, or measurement errors. If we had more time, we would add more features such as guided tutorials, practice questions, and more preset examples for students to explore. We also think it would be helpful to let users save different setups and include more calculations, such as displaying the force between multiple charges. Overall, we think our simulation is a good tool for learning the basics of electric fields and electric potential, but there is still room to improve it and make it even more useful for students.

This project resulted in an original, browser-based electrostatics simulator inspired by PhET's "Charges and Fields" tool, built from scratch as a standalone HTML file. The final version supports two modes — freely placeable point charges and a parallel plate capacitor — along with electric field vectors, traced field lines, draggable equipotential/field sensors, and a test charge that visualizes force and acceleration in real time. Together these tools let a user explore core electrostatics concepts (superposition, field direction, potential, and uniform vs. fringing fields) interactively rather than just reading about them.

Possible future improvements include adding numeric labels or a legend for field vector magnitudes, letting users set custom charge magnitudes instead of only ± 1 , adding a dielectric material option inside the capacitor to demonstrate its effect on field strength and capacitance, and exporting measurement data (like sensor readings at different plate separations) directly to a table for lab writeups.

References

Anthropic. (2026). *Claude (Sonnet 5)* [Large language model]. <https://claude.ai>

PhET Interactive Simulations. (n.d.). *Charges and fields* [Interactive simulation]. University of Colorado Boulder. https://phet.colorado.edu/sims/html/charges-and-fields/latest/charges-and-fields_all.html